# Operational Amplifiers

**Operational Amplifiers**, or **Op-amps** as they are more commonly called, are one of the basic building blocks of Analogue Electronic Circuits. *Operational amplifiers* are linear devices that have all the properties required for nearly ideal DC amplification and are therefore used extensively in signal conditioning, filtering or to perform mathematical operations such as add, subtract, integration and differentiation.

An **Operational Amplifier**, or op-amp for short, is fundamentally a voltage amplifying device designed to be used with external feedback components such as resistors and capacitors between its output and input terminals. These feedback components determine the resulting function or "operation" of the amplifier and by virtue of the different feedback configurations whether resistive, capacitive or both, the amplifier can perform a variety of different operations, giving rise to its name of "Operational Amplifier". Op-Amp ICs have become an integral part of almost all analogue circuitry. In this article we will take a look at one of the most used Op-Amp ICs: IC 741 Op Amp.

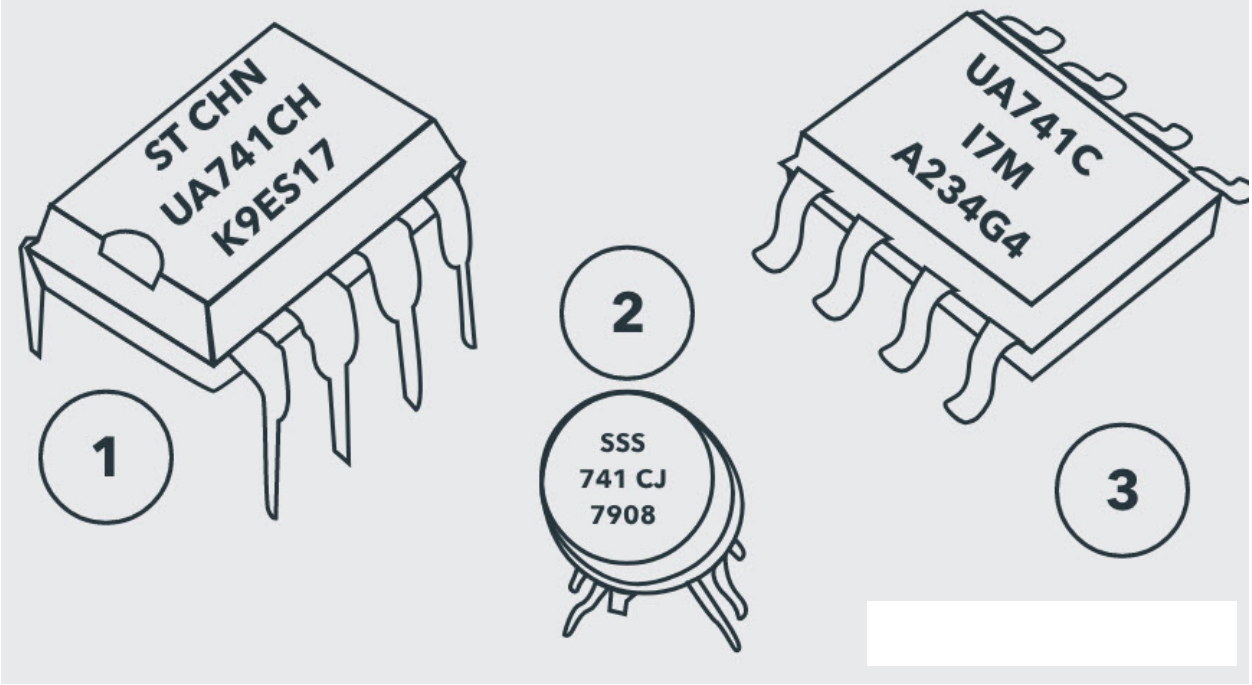
## IC 741 Op Amp (Operational Amplifier)

The 741 Op Amp IC is a monolithic integrated circuit, comprising of a general purpose Operational Amplifier. It was first manufactured by Fairchild semiconductors in the year 1963. The number 741 indicates that this operational amplifier IC has 7 functional pins, 4 pins capable of taking input and 1 output pin.

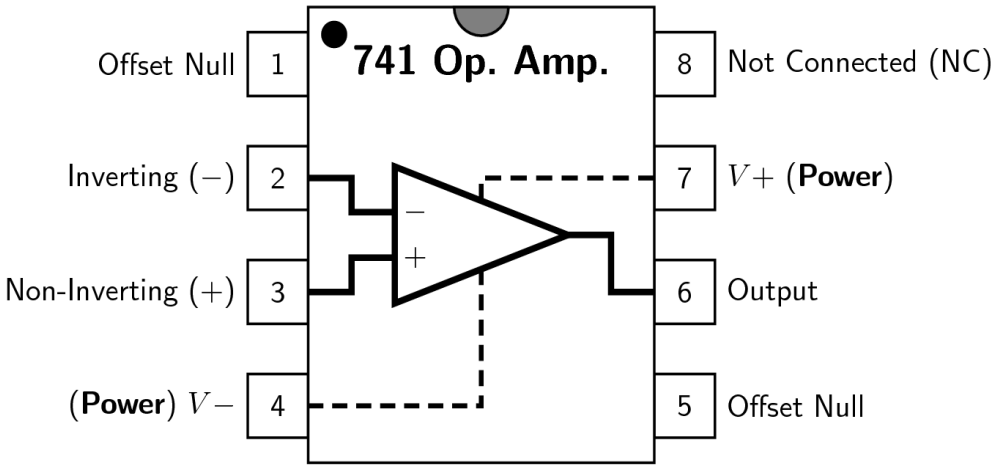
IC 741 Op Amp can provide high voltage gain and can be operated over a wide range of voltages, which makes it the best choice for use in integrators, summing amplifiers and general feedback applications. It also features short circuit protection and internal frequency

compensation circuits built in it. This Op-amp IC comes in the following form factors:

- 8 Pin DIP Package
- TO5-8 Metal can package
- 8 Pin SOIC



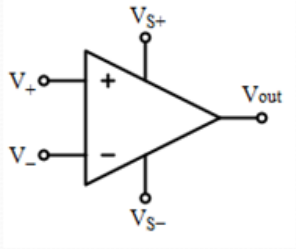

**Pinout of IC 741 and their functions**



- **Pin4 & Pin7 (Power Supply):** Pin7 is the positive voltage supply terminal and Pin4 is the negative voltage supply terminal. The 741 IC draws in power for its operation from these pins. The voltage between these two pins can be anywhere between 5V and 18V.
- **Pin6 (Output):** This is the output pin of IC 741. The voltage at this pin depends on the signals at the input pins and the feedback mechanism used. If the output is said to be high, it means that voltage at the output is equal to positive supply voltage. Similarly, if the output is said to be low, it means that voltage at the output is equal to negative supply voltage.
- **Pin2 & Pin3 (Input):** These are input pins for the IC. Pin2 is the inverting input and Pin3 is the non-inverting input. If the voltage at Pin2 is greater than the voltage at Pin3, i.e., the voltage at inverting input is higher, the output signal stays low. Similarly, if the voltage at Pin3 is greater than the voltage at Pin2, i.e., the voltage at non-inverting input is high, the output goes high.
- **Pin1 & Pin5 (Offset Null):** Because of high gain provided by 741 Op-Amp, even slight differences in voltages at the inverting and non-inverting inputs, caused due to irregularities in manufacturing process or external disturbances, can influence the output. To nullify this effect, an offset voltage can be applied at pin1 and pin5, and is usually done using a potentiometer.
- **Pin8 (N/C):** This pin is not connected to any circuit inside 741 IC. It's just a dummy lead used to fill the void space in standard 8 pin packages.

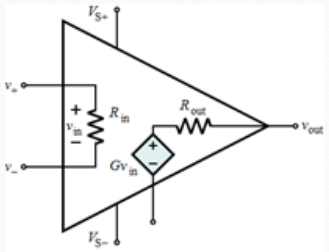
## Ideal versus Real Op-Amps

| Parameter | Ideal Op-Amp | Real Op-Amp |
|---|---|---|
| Differential Voltage Gain | ∞ | $10^5$ - $10^9$ |
| Gain Bandwidth Product (Hz) | ∞ | 1-20 MHz |
| Input Resistance (R) | ∞ | $10^6$ - $10^{12}$ Ω |
| Output Resistance (R) | 0 | 100 - 1000 Ω |




Ideal

Real




# Op-amp Parameter and Idealised Characteristic

- **Open Loop Gain, (Avo)**
  - **Infinite** – The main function of an operational amplifier is to amplify the input signal and the more open loop gain it has the better. Open-loop gain is the gain of the op-amp without positive or negative feedback and for such an amplifier the gain will be infinite but typical real values range from about 20,000 to 200,000.
- **Input impedance, ($Z_{IN}$)**

- **Infinite** – Input impedance is the ratio of input voltage to input current and is assumed to be infinite to prevent any current flowing from the source supply into the amplifiers input circuitry ( $I_{IN} = 0$ ). Real op-amps have input leakage currents from a few pico-amps to a few milli-amps.

- **Output impedance, ($Z_{OUT}$)**

- **Zero** – The output impedance of the ideal operational amplifier is assumed to be zero acting as a perfect internal voltage source with

no internal resistance so that it can supply as much current as necessary to the load. This internal resistance is effectively in series with the load thereby reducing the output voltage available to the load.

### Bandwidth, (BW)

- **Infinite** – An ideal operational amplifier has an infinite frequency response and can amplify any frequency signal from DC to the highest AC frequencies so it is therefore assumed to have an infinite bandwidth. With real op-amps, the bandwidth is limited by the Gain-Bandwidth product (GB), which is equal to the frequency where the amplifiers gain becomes unity.

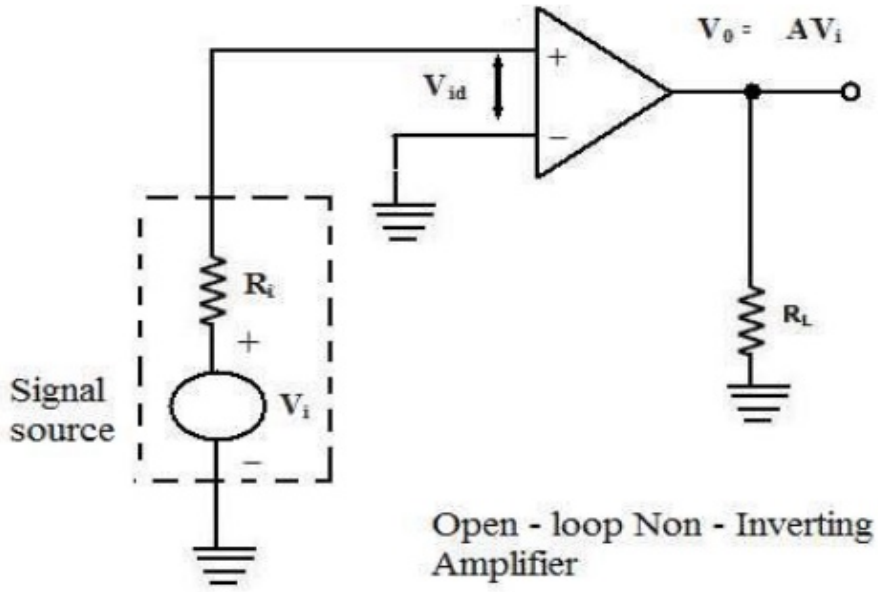
### Offset Voltage, ($V_{IO}$)

- **Zero** – The amplifiers output will be zero when the voltage difference between the inverting and the non-inverting inputs is zero, the same or when both inputs are grounded. Real op-amps have some amount of output offset voltage.

**Open and Close loop Op Amp Configuration**

**Open loop Configuration:**

The term open-loop indicates that no feedback in any form is fed to the input from the output. When connected in open – loop, the op-amp functions as a very high gain amplifier. There are three open – loop configurations of op-amp namely differential amplifier, Inverting amplifier and NON inverting amplifier. The above classification is made based on the number of inputs used and the terminal to which the input is applied. The op-amp amplifies both ac and dc input signals. Thus, the input signals can be either ac or dc voltage.




Open - loop Non - Inverting Amplifier

Figure shows the open – loop non- inverting amplifier. The input signal is applied to the non-inverting input terminal of the op-amp and the inverting input terminal is connected to the ground.

The input signal is amplified by the open – loop gain A and the output is in-phase with input signal.

In all the above open-loop configurations, only very small values of input voltages can be applied. Even for voltages levels slightly greater than zero, the output is driven into saturation, which is observed from the ideal transfer characteristics of op-amp shown in figure. Thus, when operated in the open-loop configuration, the output of the op-amp is either in negative or positive saturation, or switches between positive and negative saturation levels. This prevents the use of open – loop configuration of op-amps in linear applications.

## Limitations of Open – loop Op – amp configuration:

Firstly, in the open – loop configurations, clipping of the output waveform can occur when the output voltage exceeds the saturation

level of op-amp. This is due to the very high open – loop gain of the op-amp. This feature actually makes it possible to amplify very low frequency signal of the order of microvolt or even less, and the amplification can be achieved accurately without any distortion. However, signals of such magnitudes are susceptible to noise and the amplification for those application is almost impossible to obtain in the laboratory.

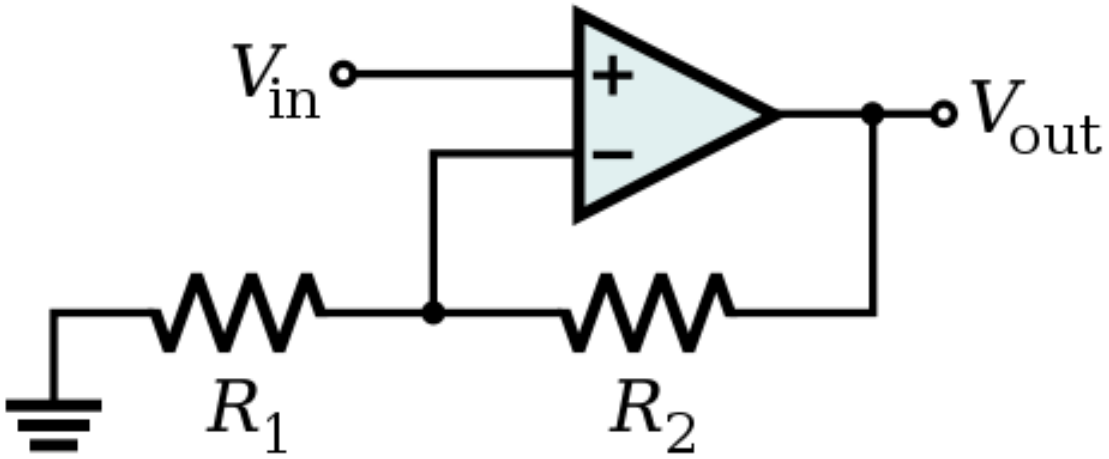
Secondly, the open – loop gain of the op – amp is not a constant and it varies with changing temperature and variations in power supply. Also, the bandwidth of most of the open- loop op amps is negligibly small. This makes the open – loop configuration of op-amp unsuitable for ac applications. The open – loop bandwidth of the widely used 741 IC is approximately 5Hz. But in almost all ac applications, the bandwidth requirement is much larger than this.

For the reason stated, the open – loop op-amp is generally not used in linear applications. However, the open – loop op amp configurations find use in certain non – linear applications such as comparators, square wave generators and astable multivibrators.

## Closed loop Configuration:

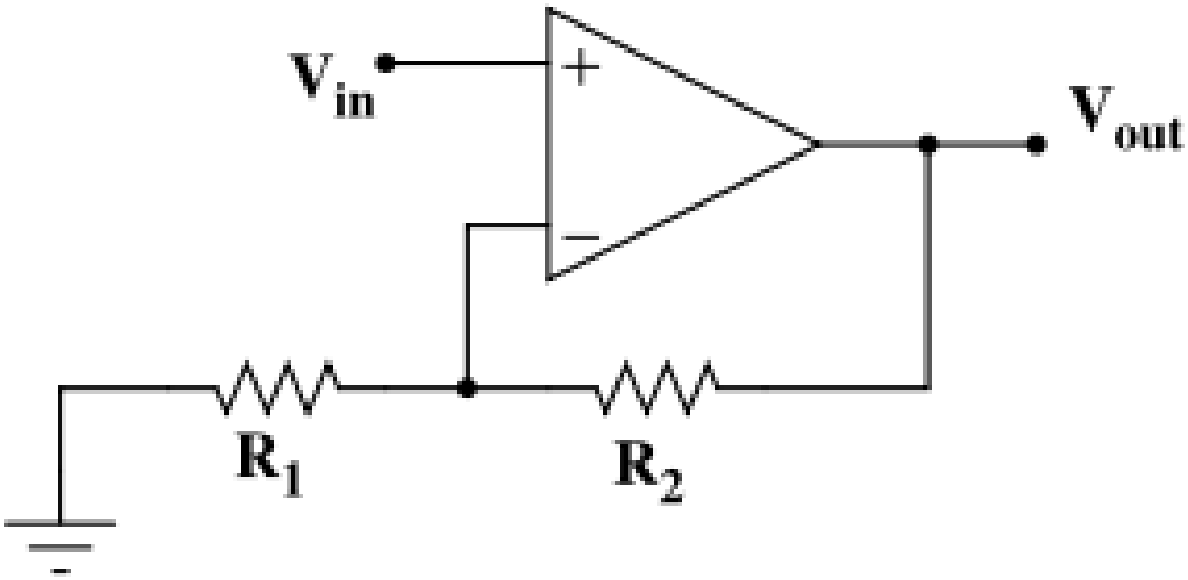
The op-amp can be effectively utilized in linear applications by providing a feedback from the output to the input, either directly or through another network. If the signal feedback is out- of-phase by 180 degree with respect to the input, then the feedback is referred to as negative feedback or degenerative feedback. Conversely, if the feedback signal is in phase with that at the input, then the feedback is referred to as positive feedback or regenerative feedback.

An op – amp that uses feedback is called a closed – loop amplifier. The most commonly used closed – loop amplifier configurations are 1. Inverting amplifier (Voltage shunt amplifier) 2. Non- Inverting amplifier (Voltage – series Amplifier)




## Applications of OPAmp:

### Non Inverting Amplifier




.**Non-inverting op-amp definition** is, when the output of an operational amplifier is in phase with an input signal then it is known as a non-inverting op-amp. In this amplifier, the input signal is applied to the +ve terminal of an operational amplifier. A non-inverting amplifier

generates an amplified output signal that is in phase with the applied input signal.

The non-inverting op-amp circuit diagram is as above. In this circuit configuration, the output voltage signal is given to the inverting terminal (-) of the operational amplifier like feedback through a resistor where another resistor is given to the ground. Here, a voltage divider with two types of resistors will provide a small fraction of the output toward the inverting pin of the operational amplifier circuit. These two resistors will provide necessary feedback to the operational amplifier.

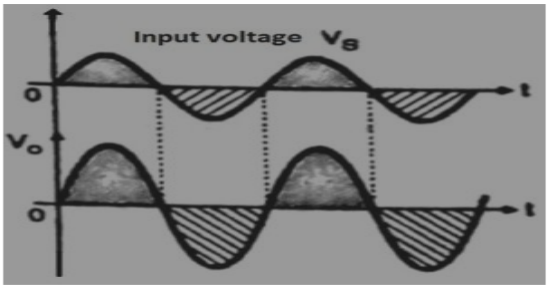
In perfect condition, the op-amp's input pin will provide maximum input impedance whereas the output pin will provide low output impedance. The output voltage is given by formula

$$V_{out} = V_{in}\left(1 + \frac{R_2}{R_1}\right)$$

## *Non Inverting Op Amp Waveform*

The input & output waveforms of non-inverting op-amp waveforms are shown below. The signal which needs to change is given to the +ve terminal of the op-amp whereas the –ve terminal is connected to GND with the help of an R1 resistor. The input (Vin) & output (Vout) voltages are within phase through each other, so their phase difference is 0 degrees or 360 degrees.

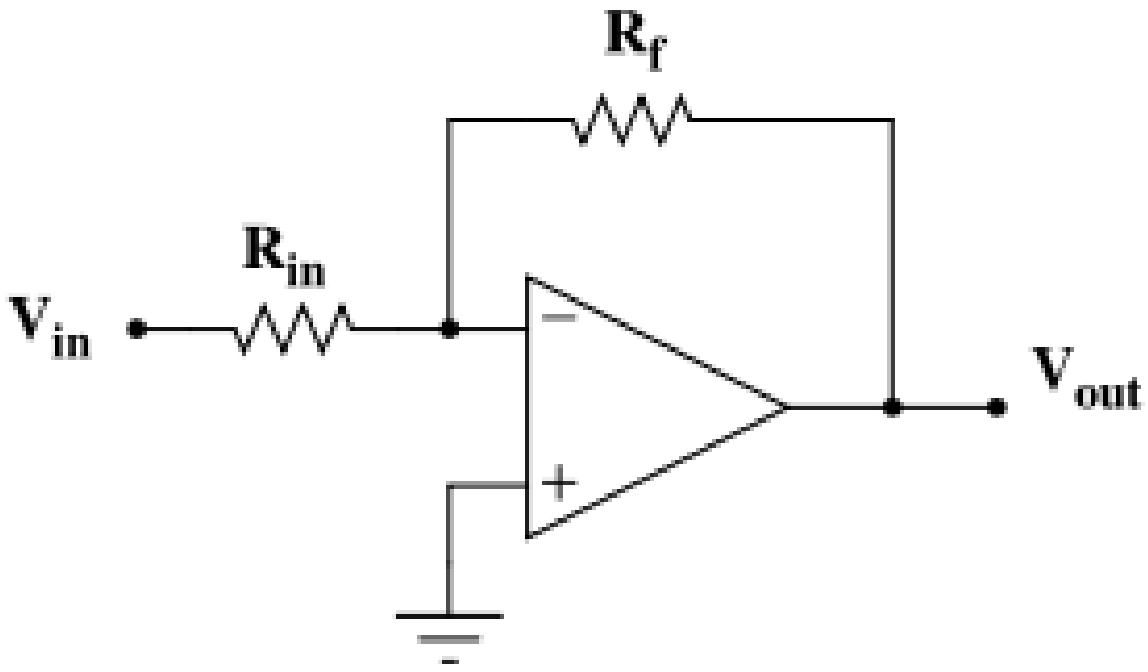
Output Waveforms



Output Waveforms

So the positive sign specifies that there is no phase shift between input & output. The voltage gain is dependent on two resistances R1 and Rf. By changing the values of the two resistances required gain can be adjusted.

## Inverting Amplifier




An **inverting op amp** is an operational amplifier circuit with an output voltage that changes in the opposite direction as the input voltage. In other words, it is out of phase by $180^o$

Consider the inverting op amp circuit shown above. The current flowing through $R_{in}$ is $I=V_{in}/R_{in}$ and since the Current Rule states that the inputs draw no current, all of that current must then flow through $R_f$.

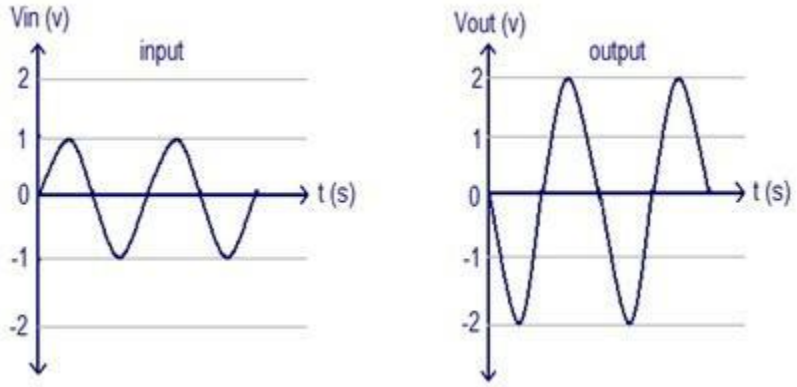
Since the inverting input is at virtual ground, the output of the inverting op amp is $V_{out} = -IR_f = -V_{in}R_f/R_{in}$.

This makes the gain of the inverting op amp circuit $-R_f/R_{in}$. The gain is negative, meaning the output is out of phase with the input.

$$V_{out} = -V_{in}(R_f/R_{in})$$

## *Inverting Op Amp Waveform*

The inverting op-amps input & output waveforms are shown below. The following waveforms can be drawn by assuming the gain of the amplifier & the sine wave is an input signal.




## Current to Voltage Converter

A **current to voltage converter** or **I to V converter** is an electronic circuit that takes current as the input and produces voltage as the output. An op-amp based current to voltage converter produces an output voltage when current is applied to its inverting terminal. The **circuit diagram** of an op-amp based current to voltage converter is shown in the following figure.

Assuming ideal Op-Amp

$A_V=\infty$ & $R_i=\infty$

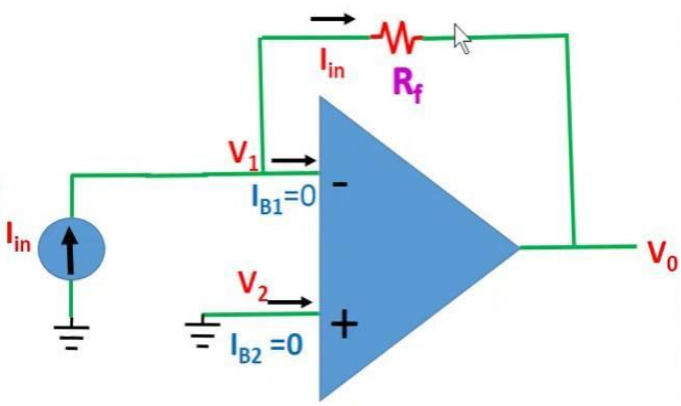
Since , $A_V=\infty$

$V_1 \approx V_2 = 0$ (By virtual ground concept)

Since, $R_i=\infty$

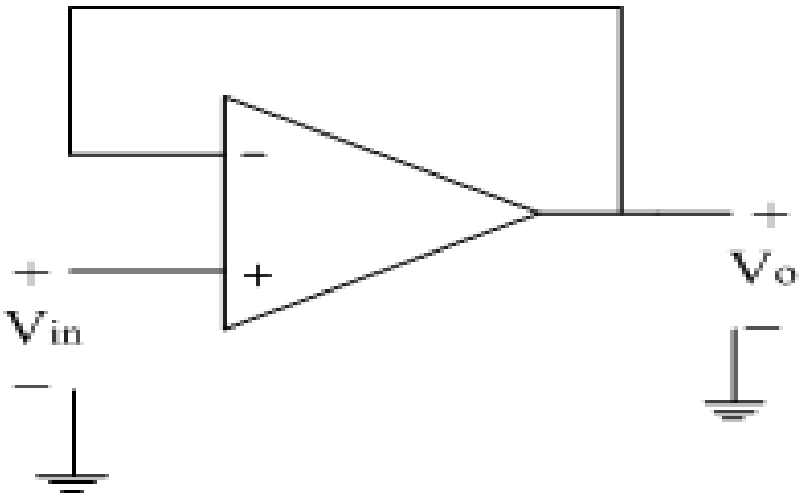
$I_{B2} = I_{B1} = 0$

$$I_{in} = \frac{V_1 - V_0}{R_f} = \frac{0 - V_0}{R_f} = \frac{-V_0}{R_f}$$




## Voltage Follower

A voltage follower (also known as a buffer amplifier, unity-gain amplifier, or isolation amplifier) is an op-amp circuit whose output voltage is equal to the input voltage. Hence a voltage follower op-amp does not amplify the input signal and has a voltage gain of 1.



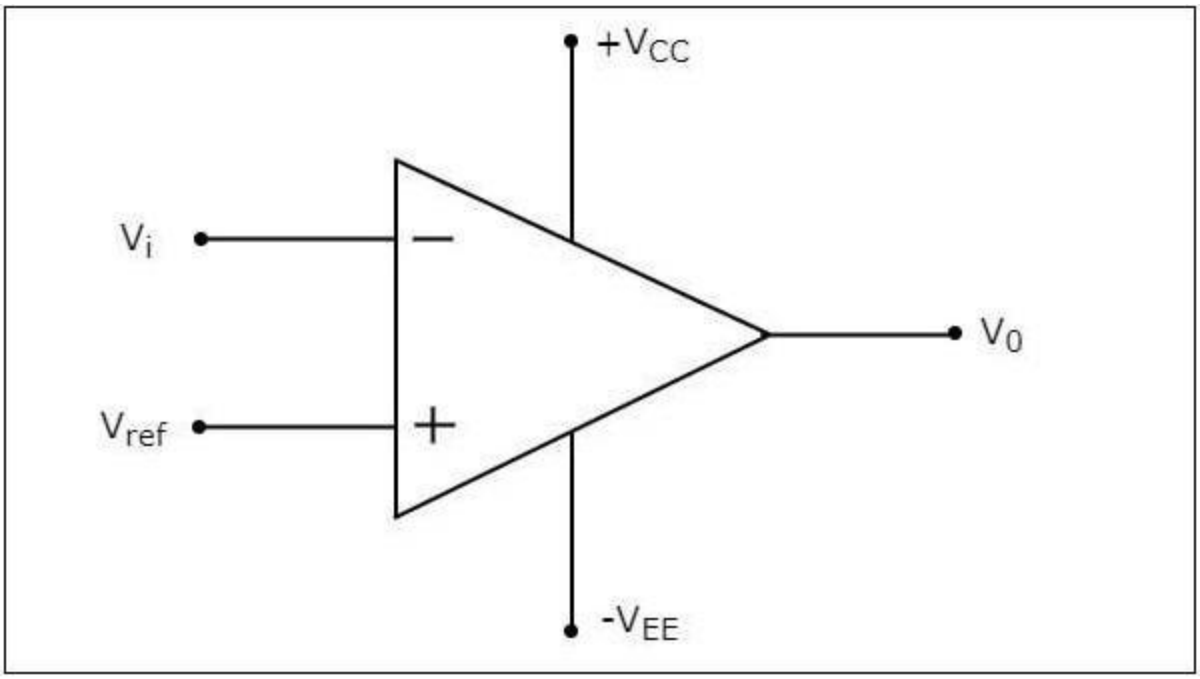

Vo= Vin

Advantages:

1) High input impedance.
2) Low output impedance.
3) High bandwidth.
4) The output follows input without phase shift.

## Comparators

A **comparator** is an electronic circuit, which compares the two inputs that are applied to it and produces an output.

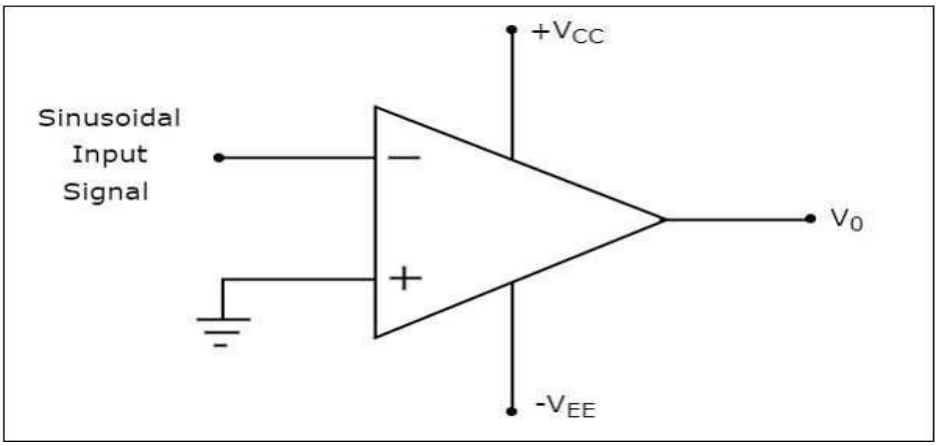



- **Types of Comparators**

  Inverting
  Non-inverting

## Inverting Comparator

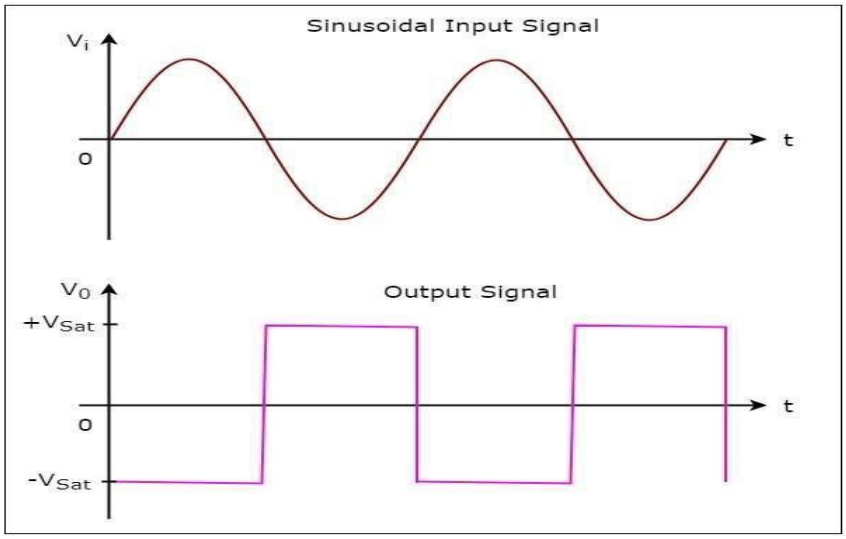
An **inverting comparator** is an op-amp based comparator for which a reference voltage is applied to its non-inverting terminal and the input voltage is applied to its inverting terminal. This comparator is called as **inverting** comparator because the input voltage, which has to be compared is applied to the inverting terminal of op-amp.




The **operation** of the inverting comparator shown above is discussed below −

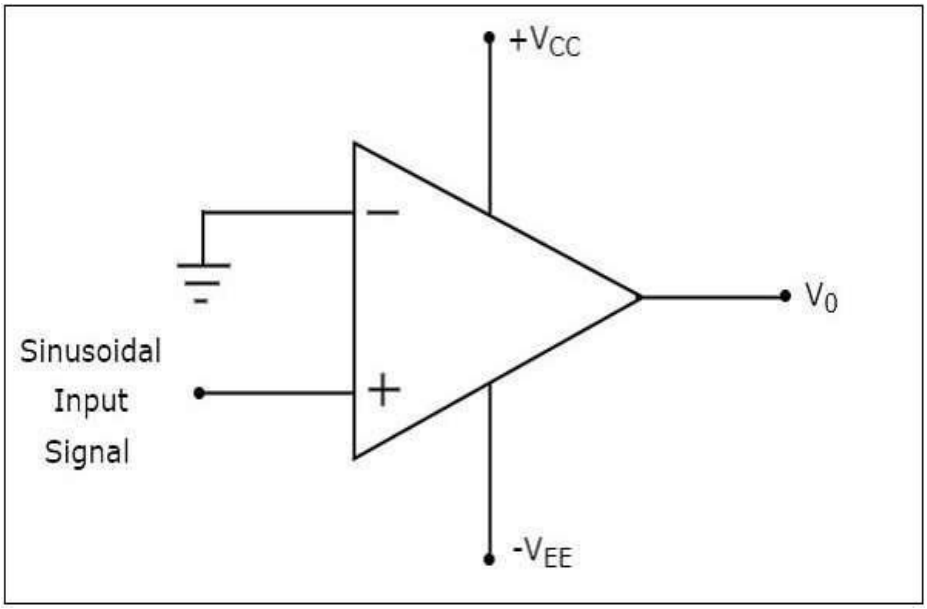
During the **positive half cycle** of the sinusoidal input signal, the voltage present at the inverting terminal of op-amp is greater than zero volts. Hence, the output value of the inverting comparator will be equal to −*Vsat* during positive half cycle of the sinusoidal input signal.

Similarly, during the **negative half cycle** of the sinusoidal input signal, the voltage present at the inverting terminal of the op-amp is less than zero volts. Hence, the output value of the inverting comparator will be equal to +*Vsat* during negative half cycle of the sinusoidal input signal.




## Non-inverting comparator

A non-inverting comparator is an op-amp based comparator for which a reference voltage is applied to its inverting terminal and the input voltage is applied to its non-inverting terminal. This op-amp based comparator is called as **non-inverting** comparator because the input voltage, which has to be compared is applied to the non-inverting terminal of the op-amp.



The **operation** of a non-inverting comparator is explained below

During the **positive half cycle** of the sinusoidal input signal, the voltage present at the non-inverting terminal of op-amp is greater than zero volts. Hence, the output value of a non-inverting comparator will be equal to $+Vsat$ during the positive half cycle of the sinusoidal input signal.

Similarly, during the **negative half cycle** of the sinusoidal input signal, the voltage present at the non-inverting terminal of op-amp is less than zero volts. Hence, the output value of non-inverting comparator will be equal to $-Vsat$ during the negative half cycle of the sinusoidal input signal.

